

## 5.2.2 Properties of the Definite Integral

Properties of the Definite Integral

**Theorem:** Suppose that  $f$  and  $g$  are integrable functions on  $[a, b]$ . Then we have that:

$$1. \int_a^a f(x) dx = 0$$

$$2. \int_a^b f(x) dx = -\int_b^a f(x) dx$$

$$3. \int_a^b c dx = c(b-a), \text{ where } c \text{ is any constant.}$$

$$4. \int_a^b c \cdot f(x) dx = c \cdot \int_a^b f(x) dx, \text{ where } c \text{ is any constant.}$$

$$5. \int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$6. \int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

$$7. \int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

**Example 1:** Suppose that  $f$  is continuous at all real number and that  $\int_2^6 f(x) dx = 1$ . What is the value of

$$\int_6^2 [2 - 3f(x)] dx ?$$

**Example 2:** If it is known that  $\int_0^{10} f(x) dx = 17$  and  $\int_0^8 f(x) dx = 12$ , find  $\int_8^{10} f(x) dx$ .

**Example 3:** Suppose that  $f$  is continuous at all real number and that  $\int_1^3 f(x) dx = 4$  and  $\int_1^5 f(x) dx = 13$ . What is the value of  $\int_3^5 [2f(x) + 10] dx$ ?

**Theorem (Comparison Properties of the Integral):** Suppose that  $f$  and  $g$  are integrable functions on  $[a, b]$  and  $m$  and  $M$  are real numbers. Then we have that:

8. If  $f(x) \geq 0$  for  $a \leq x \leq b$ , then  $\int_a^b f(x) dx \geq 0$ .

9. If  $f(x) \geq g(x)$  for  $a \leq x \leq b$ , then  $\int_a^b f(x) dx \geq \int_a^b g(x) dx$ .

10. If  $m \leq f(x) \leq M$  for  $a \leq x \leq b$ , then  $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$ .

*Note: Property 10 is very useful for estimating the value of difficult integrals.*

**Example 4:** Use the properties of integrals to estimate the integral  $\int_{-1}^1 \frac{1}{x^2 + 1} dx$ .

**Example 5:** Use the properties of integrals to determine whether the value of  $\int_{-2}^2 \frac{x^3 - 9}{|x| + 1} dx$  is positive or negative.

**Example 6:** Evaluate the integral  $\int_{-3}^0 (1 + \sqrt{9 - x^2}) dx$  by interpreting it in terms of areas.